

```
# Install spaCy (run in terminal or notebook)
!pip install spacy

# Download the English language model for spaCy
!python -m spacy download en_core_web_sm

# Import the spaCy library
import spacy

# Load the downloaded English language model
nlp = spacy.load("en_core_web_sm")

# Verify that the model was loaded successfully
print("Model loaded successfully!")
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl: 13.8/12.8 MB 67.3 MB/s eta 0:00:00
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.
Model loaded successfully!

```
# Define a list of sentences to be processed
sentences = [
    "The cat sits on the mat.",
    "She is reading a book.",
    "The quick brown fox jumps over the lazy dog.",
    "He bought a new car yesterday.",
    "They are playing in the park."
]
```

```
# Iterate through each sentence in the list
for sent in sentences:
    # Process the sentence with the spaCy NLP model
    doc = nlp(sent)
    # Print the original sentence
    print("\nSentence:", sent)
```

```
# Iterate through each token (word) in the processed document
for token in doc:
    # Print the token's text, part-of-speech tag, and dependency relation
    print(token.text, token.pos_, token.dep_)
```

```
Sentence: The cat sits on the mat.
The DET det
cat NOUN nsubj
sits VERB ROOT
on ADP prep
the DET det
mat NOUN pobj
. PUNCT punct
```

```
Sentence: She is reading a book.
She PRON nsubj
is AUX aux
reading VERB ROOT
a DET det
book NOUN dobj
. PUNCT punct
```

```
Sentence: The quick brown fox jumps over the lazy dog.
The DET det
quick ADJ amod
brown ADJ amod
fox NOUN nsubj
jumps VERB ROOT
over ADP prep
the DET det
lazy ADJ amod
dog NOUN pobj
. PUNCT punct
```

```
Sentence: He bought a new car yesterday.
He PRON nsubj
bought VERB ROOT
a DET det
new ADJ amod
car NOUN dobj
yesterday NOUN npadvmod
. PUNCT punct
```

```
Sentence: They are playing in the park.
They PRON nsubj
are AUX aux
playing VERB ROOT
in ADP prep
the DET det
park NOUN pobj
. PUNCT punct
```

```
# Import the pandas library for data manipulation
import pandas as pd

# Iterate through each sentence in the list
for sent in sentences:
    # Process the sentence with the spaCy NLP model
    doc = nlp(sent)

    # Initialize an empty list to store token data
    data = []
    # Iterate through each token in the processed document
    for token in doc:
        # Append a list containing token's text, lemma, POS, dependency, and head to the data list
        data.append([
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])

    # Create a pandas DataFrame from the collected token data
    df = pd.DataFrame(data, columns=["Token", "Lemma", "POS", "Dependency", "Head"])

    # Print the original sentence
    print("\nSentence:", sent)
    # Print the DataFrame containing token information
    print(df)
```

```
Sentence: The cat sits on the mat.
Token Lemma    POS Dependency Head
0 The the DET det cat
```

```

1  cat  cat  NOUN    nsubj  sits
2  sits sit  VERB     ROOT   sits
3  on   on   ADP      prep   sits
4  the  the  DET      det    mat
5  mat  mat  NOUN    pobj   on
6  .    .    PUNCT    punct  sits

```

Sentence: She is reading a book.

	Token	Lemma	POS	Dependency	Head
0	She	she	PRON	nsubj	reading
1	is	be	AUX	aux	reading
2	reading	read	VERB	ROOT	reading
3	a	a	DET	det	book
4	book	book	NOUN	dobj	reading
5	.	.	PUNCT	punct	reading

Sentence: The quick brown fox jumps over the lazy dog.

	Token	Lemma	POS	Dependency	Head
0	The	the	DET	det	fox
1	quick	quick	ADJ	amod	fox
2	brown	brown	ADJ	amod	fox
3	fox	fox	NOUN	nsubj	jumps
4	jumps	jump	VERB	ROOT	jumps
5	over	over	ADP	prep	jumps
6	the	the	DET	det	dog
7	lazy	lazy	ADJ	amod	dog
8	dog	dog	NOUN	pobj	over
9	.	.	PUNCT	punct	jumps

Sentence: He bought a new car yesterday.

	Token	Lemma	POS	Dependency	Head
0	He	he	PRON	nsubj	bought
1	bought	buy	VERB	ROOT	bought
2	a	a	DET	det	car
3	new	new	ADJ	amod	car
4	car	car	NOUN	dobj	bought
5	yesterday	yesterday	NOUN	npadvmod	bought
6	.	.	PUNCT	punct	bought

Sentence: They are playing in the park.

	Token	Lemma	POS	Dependency	Head
0	They	they	PRON	nsubj	playing
1	are	be	AUX	aux	playing
2	playing	play	VERB	ROOT	playing
3	in	in	ADP	prep	playing
4	the	the	DET	det	park
5	park	park	NOUN	pobj	in
6	.	.	PUNCT	punct	playing

```

# Iterate through each sentence in the list
for sent in sentences:
    # Process the sentence with the spaCy NLP model
    doc = nlp(sent)

    # Initialize variables for subject, verb, and object
    subject = ""
    verb = ""
    obj = ""

    # Iterate through each token in the processed document
    for token in doc:
        # If the token's dependency relation is 'nsubj' (nominal subject), assign it to 'subject'
        if token.dep_ == "nsubj":
            subject = token.text
        # If the token's dependency relation is 'ROOT', assign it to 'verb'
        if token.dep_ == "ROOT":
            verb = token.text
        # If the token's dependency relation is 'dobj' (direct object), assign it to 'obj'
        if token.dep_ == "dobj":
            obj = token.text

    # Print the original sentence
    print("\nSentence:", sent)
    # Print the extracted subject
    print("Subject:", subject)
    # Print the extracted verb
    print("Verb:", verb)
    # Print the extracted object
    print("Object:", obj)

```

Sentence: The cat sits on the mat.
 Subject: cat
 Verb: sits
 Object:

Sentence: She is reading a book.
Subject: She
Verb: reading
Object: book

Sentence: The quick brown fox jumps over the lazy dog.
Subject: fox
Verb: jumps
Object:

Sentence: He bought a new car yesterday.
Subject: He
Verb: bought
Object: car

Sentence: They are playing in the park.
Subject: They
Verb: playing
Object:

```
# Import displacy for visualizing dependency parses
from spacy import displacy

# Iterate through each sentence in the list
for sent in sentences:
    # Process the sentence with the spaCy NLP model
    doc = nlp(sent)
    # Render the dependency parse tree for the document in Jupyter Notebook
    displacy.render(doc, style="dep", jupyter=True)
```

